

Software overview

Language	: C++ (also supports C)
Interfaces	: Python, Java and MATLAB/OCTAVE
Wrappers	: C#, Perl, Ch and Ruby
Operating system	: Desktop - Windows, Linux, OS X, FreeBSD, NetBSD and OpenBSD Mobile - Android, iOS, Maemo and BlackBerry 10
Release sources	: SourceForge, GitHub
Built using	: C Make
Licence	: 3-clause Berkeley Software Distribution (BSD) licence
Website	: opencv.org
Corporate users	: Google, Yahoo, Microsoft, Intel, IBM, Sony, Honda and Toyota, and start-ups such as Applied Minds, VideoSurf and Zeitera



Fig. 2: Creating sketch effect by detecting edges (Image courtesy: www.bogotobogo.com)

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Segmentation, edge detection and morphology. With this tool, the image under processing can be partitioned into multiple segments by using the thresholding approach. Also, points, lines and edges in an image can be easily detected by simply applying different order derivatives to the image. Fig. 2 shows what an edge-detected image looks like.

Morphological operations process the image based on shapes or templates called structuring elements, and we can apply image pyramids to either upsize or downsize the image.

Histogram. Analyse your image by obtaining the image histogram and studying the intensity distribution. The tool scans the entire image, tabulates different intensities and increases the corresponding intensity count as it scans the image, simultaneously. Fig. 3 shows a sample histogram.

We can use this information, which is generally seen as a graph, to fix threshold limits and decide the

pixel adjustments to be made to the image. The histogram can also be equalised to improve image contrast.

Transformations. Convert the image from the existing spatial domain to corresponding frequency domain by applying Fourier transform. To find imperfect instances of objects, like circles and other classes of shapes in your image, apply Hough transform. The image can also be remapped by moving existing pixels to a new image.

The complex math and a little beyond. Statistical evaluation using expectation-maximisation algorithm, k-nearest neighbour algorithm, random forest and support vector machine is made easy by simply coding it into the tool and then applying the same on the image.

Decision tree learning can be used to predict an image's target value by mapping observations. Multi-dimensional scaling, naive bayers classifiers and gradient boosting help you work with the images better. Image analysis can even be extended to the level of neural networks, converting neuron connections to numeric weights that can be worked with.

If you want to make your own videos with existing images, you can interact with and manipulate the real world around using augmented-reality feature.

The added modules. We can create three dimensional (3D) reality from two dimensional (2D) images using Calib3d. Point detectors, descriptors and matching framework options can

be found in Feature2d. Algorithms to perform motion extraction, feature tracking and foreground extractions on the video stream are provided in the video module.

Graphical processing. An exclusive graphical processing unit (GPU) module helps squeeze out every little computation power from the system by using the power of the video card to run OpenCV algorithms. Based on compute unified device architecture (CUDA), the module provides GPU acceleration and helps developers run more accurate and sophisticated OpenCV algorithms in real-time on higher-resolution images, while consuming less power.

Added support to OpenCL in the library enables the codes to take advantage of the heterogeneous hardware, in particular utilise the potential of discrete and integrated GPUs.

Graphical user interface (GUI). The built-in GUI allows easy access to the functionalities of OpenCV. We can add track bars to applications and also read videos or create new ones. There are also methods to perform similarity measurements to check if two videos or images are copies of each other.

Pointers from the latest release

OpenCV 3.0 gold release, as it is popularly called, has a transparent application program interface, which is a GPU-acceleration layer using OpenCL, introduced with support from AMD and Intel. With access to a sub-set of Intel's integrated performance primitives, improved Python and Java bindings, new MATLAB bindings and accelerated functions using NEON intrinsics, it promises to be the most functional and fastest OpenCV ever.

Projects from Google Summer of Code programme have been integrated into this release. Biologically-inspired vision module, DAISY features, LATCH descriptor, improved BRIEF and image registration module are contributions from communities.